

GO_Biological_Process_2023 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
NADH Dehydrogenase Complex Assembly (GO: -0.36633831	48	1.728e-18	4.655e-15		TMEM126A:5 NDUFS3:30 NDUFB8:45 NDUFB10:57 TMEM126B:65 NDUFS5:72
Mitochondrial Respiratory Chain Complex -0.38633831	48	1.728e-18	4.655e-15		TMEM126A:5 NDUFS3:30 NDUFB8:45 NDUFB10:57 TMEM126B:65 NDUFS5:72
Mitochondrial Respiratory Chain Complex -0.28129839	77	1.565e-17	2.809e-14		TMEM126A:5 NDUFS3:30 NDUFB8:45 NDUFB10:57 TMEM126B:65 NDUFS5:72
Mitochondrial Gene Expression (GO:014005 -0.23480638	99	7.717e-16	1.039e-12		MRPL42:40 MRPL37:95 FASTKD1:121 MRPS30:162 PTCOD3:163 MRPL24:172
Monatomic Cation Transmembrane Transport 0.14014831	269	3.533e-15	3.807e-12		ATP6V0A2:7 TRPV6:25 KCNQ3:45 ANO6:48 ATP2A1:91 SLC30A10:118
Inorganic Cation Transmembrane Transport 0.13751582	272	8.148e-15	7.316e-12		ATP6V0A2:7 TRPV6:25 KCNQ3:45 ANO6:48 ATP2A1:91 SLC30A10:118
Translation (GO:0006412) -0.17113827	167	2.824e-14	2.173e-11		MRPL42:40 MRPL37:95 MRPS30:162 PTCOD3:163 MRPL24:172 RARS2:192
Mitochondrial Translation (GO:0032543) -0.21226849	95	9.495e-13	6.394e-10		MRPL42:40 MRPL37:95 MRPS30:162 PTCOD3:163 MRPL24:172 RARS2:192
Oxidative Phosphorylation (GO:0006119) -0.27719912	54	1.925e-12	1.152e-09		NDUFS3:30 NDUFB8:45 NDUFB10:57 NDUFS5:72 NDUFA11:78 ATP5PD:118
Proton Motive Force-Driven Mitochondrial Transp -0.28974253	47	4.560e-12	3.534e-09		NDUFS3:30 NDUFB8:45 NDUFB10:57 NDUFS5:72 NDUFA11:78 ATP5PD:118
Aerobic Respiration (GO:000960) -0.26354689	53	3.332e-11	1.632e-08		NDUFS3:30 NDUFB8:45 NDUFB10:57 NDUFS5:72 NDUFA11:78 NDUFA8:245
Proton Motive Force-Driven ATP Synthesis -0.26757892	51	3.979e-11	1.786e-08		NDUFS3:30 NDUFB8:45 NDUFB10:57 NDUFS5:72 NDUFA11:78 ATP5PD:118
Mitochondrial Electron Transport, NADH T -0.33967433	31	6.051e-11	2.508e-08		NDUFS3:30 NDUFB8:45 NDUFB10:57 NDUFS5:72 NDUFA8:245 NDUFA9:279
Mitochondrial ATP Synthesis Coupled Elec -0.25677016	52	1.575e-10	6.062e-08		NDUFS3:30 NDUFB8:45 NDUFB10:57 NDUFS5:72 NDUFA8:245 NDUFA12:259
Aerobic Electron Transport Chain (GO:001 -0.24970207	52	4.863e-10	1.747e-07		NDUFS3:30 NDUFB8:45 NDUFB10:57 NDUFS5:72 NDUFA8:245 NDUFA9:279
Peptide Biosynthetic Process (GO:0043043 -0.17104180	99	4.353e-09	1.379e-06		MRPL42:40 MRPL37:95 MRPL24:172 RPS1:739 RPS23:201 ZAR1:261
Regulation Of Transcription By RNA Polym -0.04657229	1527	4.113e-09	1.379e-06		BRCA1:11 ZSCAN5B:34 N4BP2L:237 KLF17:42 ZSCAN20:60 ZFP90:74
Cellular Respiration (GO:0045333) -0.19778894	69	1.392e-08	4.166e-06		NDUFS3:30 NDUFB8:45 NDUFB10:57 NDUFS5:72 NDUFA11:78 NDUFA8:245
Regulation Of DNA-templated Transcrip tio -0.04458481	1474	2.948e-08	8.359e-06		BRCA1:11 ZMYND11:24 ZSCAN5B:34 KLF17:42 ZSCAN20:60 ZFP90:74
Potassium Ion Transmembrane Transport (G 0.14033469	130	3.546e-08	9.552e-06		KCNQ3:45 KCNF1:158 KCNK18:188 TMEM175:289 KCNQ5:350 KCNG4:427
Metal Ion Transport (GO:0030001) 0.12550412	148	1.477e-07	3.789e-05		TRPV6:25 ATP2A1:91 SLC5A8:145 KCNF1:158 SCN1G:162 KCNK18:188
IRNA Modification (GO:0006400) -0.18679251	66	1.590e-07	3.894e-05		THADA:106 TRMT13:173 PUS10:391 TRMT10A:402 TRMT6:491 OSGEPL1:528
IRNA Methylation (GO:0030488) -0.24643963	36	3.153e-07	7.385e-05		TRMT10C:104 THADA:106 TRMT13:173 TRMT10A:402 TRMT6:491 METTL6:627
Mitochondrial RNA Metabolic Process (GO: -0.33249627	19	5.268e-07	1.182e-04		PNPT1:96 FASTKD1:121 POLRMT:273 FASTMD5:330 FASTKD2:330 MRPL24:172
Potassium Ion Transport (GO:0006813) 0.12748738	116	2.15e-06	4.773e-04		KCNQ3:45 KCNF1:158 KCNK18:188 TMEM175:289 KCNQ5:350 KCNG4:427
RNA Methylation (GO:001510) -0.17465289	57	5.202e-06	1.076e-03		TRMT10C:104 THADA:106 TRMT13:173 ZCCHC4:219 METTL6:1301 SNRPD3:351
Calcium Ion Transport (GO:0006816) 0.13059703	102	5.392e-06	1.076e-03		TRPV6:25 CRACR2B:31 ANO6:48 ATP2A1:91 TRPM5:198 MS4A1:351
Sodium Ion Transmembrane Transport (GO:0 0.14671047	77	8.799e-06	1.693e-03		SCNN1G:162 SLC6A8:573 SLC9A4:633 SLC6A11:671 ATP1A1:727 SCNN1A:963
Inorganic Cation Import Across Plasma Me -0.26856442	95	1.471e-05	2.641e-03		TRPV6:25 ACANA18:88 SCN9A:162 CACNA11:224 CLU:426 SLC9A4:633
Macromolecule Biosynthetic Process (GO:0 -0.11401510	122	1.428e-05	2.641e-03		MRPL42:40 MRPL37:95 MRPL24:172 RPS1:739 RPS23:201 ZAR1:260
piRNA Processing (GO:0034587) 0.30135136	17	1.701e-05	2.957e-03		MOV10L1:14 GPAT2:41 PIWIL1:106 MAEL:189 ASZ1:239 HENMT1:278
Monatomic Ion Transport (GO:0006811) 0.12257782	103	1.784e-05	3.004e-03		ANO6:48 SLC5A8:145 SLC17A4:210 SLC4A1:300 SLC1A1:381 SLC26A4:401
Mitochondrial RNA Processing (GO:0000963 -0.40620085	9	2.448e-05	3.996e-03		SUPV3L1:32 PNPT1:96 TRMT10C:104 FASTKD1:121 FASTKD5:330 FASTKD2:739
Positive Regulation Of Cytosolic Calcium 0.23673100	25	4.208e-05	6.667e-03		GPR35:272 OPR1:365 GPR20:411 F2AR1:561 GPR17:569 GRM1:750
Intermediate Filament Organization (GO:0 0.16739041	48	6.098e-05	9.386e-03		KRT36:488 PKP1:501 BFSP2:507 KRT85:511 KRT17:523 KRT72:611
Learning (GO:0007612) 0.18293494	37	1.189e-04	1.779e-02		SHANK3:94 SYNJ1:254 SHANK2:311 TH50:50 NPAS4:982 SLC8A2:1252
Positive Regulation Of Cellular Componen -0.11255978	96	1.420e-04	2.068e-02		ASAP1:5 SLC18A3:84 SHANK3:94 FRMPD4:107 WAR51:128 ADD1:157
DNA-templated DNA Replication (GO:000626 -0.13653500	64	1.611e-04	2.284e-02		DNA2:63 ORC3:236 LRWD1:256 ORC1:268 ORC4:804 TOPBP1:842
Neurotransmitter Transport (GO:0006836) 0.15637970	47	2.104e-04	2.906e-02		SLC18A3:84 SYNJ1:254 SLC1A1:381 STXBP3:820 SLC18A2:905 HRH3:1143
Regulation Of Protein Dephosphorylation 0.20163455	28	2.230e-04	3.003e-02		PPP1R2C:28 PPP6R1:389 SMPD1:496 PTPA:539 CDH5:673 TGFBI:819

EnrichmentHsSymbolsFile2 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
WP_MITOCHONDRIAL_COMPLEX_I_ALIGNMENT_MODE	-0.35116368	47	8.355e-17	5.420e-13	NDUFS3:30 NDUFB8:45 NDUFB10:57 TMEM126B:65 NDUFS5:72 NDUFAF4:132
REACTOME_COMPLEX_I_BIOGENESIS	-0.33944645	48	4.182e-16	1.356e-12	NDUFS3:30 NDUFB8:45 NDUFB10:57 TMEM126B:65 NDUFS5:72 NDUFA11:78
REACTOME_RESPIRATORY_ELECTRON_TRANSPORT_	-0.23295304	99	1.224e-15	2.569e-12	LRPPRC:1 NDUFS3:30 NDUFB8:45 NDUFB10:57 TMEM126B:65 NDUFS5:72
REACTOME_RESPIRATORY_ELECTRON_TRANSPORT_	-0.25956442	79	1.584e-15	2.569e-12	LRPPRC:1 NDUFS3:30 NDUFB8:45 NDUFB10:57 TMEM126B:65 NDUFS5:72
FISCHER_DREAM_TARGETS	-0.08121102	803	9.072e-15	1.154e-11	KIF18B:3 BRCA1:11 H2BC21:15 FAM161A:26 HROB:53 ARL13B:56
REACTOME_TRANSPORT_OF_SMALL_MOLECULES	0.09129539	624	1.068e-14	1.154e-11	ATP6V0A2:7 TRPV6:25 ANO6:48 MTPP:55 ATP2A1:91 SLC30A10:118
REACTOME_MITOCHONDRIAL_TRANSLATION	-0.22883178	91	4.764e-14	4.415e-11	MRPS35:29 MRPL42:40 MRPL37:95 MRPS30:162 PTCOD3:163 MRPL24:172
REACTOME_TRANSLATION	-0.13697878	216	4.443e-12	3.602e-09	MRPS35:29 MRPL42:40 SARS2:47 MRPL37:95 MRPS30:162 PTCOD3:163
REACTOME_THE_CITRIC_ACID_CYCLE_AND_R	-0.16432916	146	7.704e-12	4.997e-09	LRPPRC:1 NDUFS3:30 NDUFB8:45 NDUFB10:57 TMEM126B:65 NDUFS5:72
WP_OXIDATIVE_PHOSPHORYLATION	-0.29183866	46	7.609e-12	4.997e-09	NDUFS3:30 NDUFB8:45 NDUFB10:57 NDUFS5:72 NDUFA11:78 ATP5PD:118
JOHNSTONE_PARV8_TARGETS_3_DN	-0.07428848	690	4.120e-11	2.430e-08	N4BP2L2:37 NDUFS5:72 CEP350:76 NDUFA11:78 ZWINT:100 RTKN2:102
YOSHIMURA_MAPK8_TARGETS_UP	0.05763199	1063	3.612e-10	1.952e-07	RGS9:3 EPN1:34 KKNQ3:45 H1L-6:80 GABRB3:104 ADRB1:112
GEORGES_TARGETS_OF_MIR192_AND_MIR215	-0.06722744	760	4.044e-10	2.018e-07	BRCA1:11 ZMYND11:24 LIMCH1:36 POLQ:98 RTKN2:102 DLGAP5:134
WP_ELECTRON_TRANSPORT_CHAIN_OXP_HOS_SYSTE	-0.21169532	72	5.396e-10	2.500e-07	NDUFS3:30 NDUFB8:45 NDUFB10:57 NDUFS5:72 ATP5PD:118 ATP5PB:225
DODD_NASOPHARYNGEAL_CARCINOMA_DN	-0.05549732	1111	7.197e-10	3.112e-07	LRPPRC:1 KIF18B:3 BRCA1:11 LRRC40:12 GHR:23 VCAN:31
NIKOLSKY_BREAST_CANCER_7P22_AMPLICON	0.28153888	38	1.924e-09	7.780e-07	BRAT1:234 MAD1L1:292 AP5Z1:332 SNTN:348 SUN1:374 PAPOLB:405
MARTENS_TRETINOIN_RESPONSE_UP	0.06978518	643	2.039e-09	7.780e-07	GJB4:13 SCIMP:40 CBARP:109 ARX:113 REGA:163 ADAMTS8:173
REACTOME_SLC_MEDIATED_TRANSMEMBRANE_TRAN	0.11380240	232	2.569e-09	9.258e-07	SLC30A10:118 SLC5A8:145 SLC38A3:291 SLC4A1:300 SLC2A9:320 SLC22A8:358
REACTOME_KERATINIZATION	0.18846053	83	3.015e-09	1.030e-06	KLK13:77 ST14:240 KRT36:488 PKP1:501 KRT85:511 KRT17:523
RODRIGUES_THYROID_CARCINOMA_POORLY_DIFFE	-0.07451415	533	4.904e-09	1.590e-06	LRPPRC:1 BRCA1:11 MRPL42:40 SSX2IP:43 DNAJC2:75 MRPL37:95
HAMAL_APOPTOSIS_VIA_TRAIL_UP	-0.07195731	567	5.881e-09	1.817e-06	BRCA1:11 LRRC40:12 N4BP2L2:37 PTNP1:317 CEP350:76 POLQ:98
KEGG_OXIDATIVE_PHOSPHORYLATION	-0.17291789	92	1.020e-08	3.009e-06	NDUFS3:30 NDUFB8:45 NDUFB10:57 NDUFS5:72 NDUFA11:78 ATP5PD:118
SHEN_SMARCA2_TARGETS_UP	-0.07174371	360	1.226e-08	3.457e-06	SCFD1:7 ZMYND11:24 LIMCH1:36 TMEM126B:65 CEP350:76 CENPC:128
REACTOME_EPIGENETIC_REGULATION_OF_GENE_E	-0.16732064	96	1.511e-08	4.083e-06	H2BC21:15 TET1:87 TET2:371 BAZ2A:390 RRP8:423 PHF19:429
MARSON_BOUND_BY_E2F4_UNSTIMULATED	-0.06714664	600	2.401e-08	6.230e-06	KIF18B:3 LRRC40:12 SARS2:47 HROB:53 SCFD2:59 POLQ:98
MOOTHA_VOXPHOS	-0.19357318	69	2.747e-08	6.854e-06	NDUFS3:30 NDUFB8:45 NDUFS5:72 ATP5PD:118 ATP5PB:225 NDUFA8:245
REACTOME_FORMATION_OF_THE_CORNIFIED_ENVE	0.18030907	76	5.604e-08	1.332e-05	KLK13:77 ST14:240 KRT36:488 PKP1:501 KRT85:511 KRT17:523
REACTOME_SENSORIAL_PERCEPTION	0.09259615	292	5.748e-08	1.332e-05	RGS9:3 ORBD4:93 SCN1G:162 TRPM5:198 ANO2:293 R1P0R:2310
MILL_PSEUDOPODIA_HAPTOTAXIS_UP	-0.07846646	408	6.167e-08	1.379e-05	ZMYND11:24 MAPKBP1:52 PIK3C2A:171 ORC3:236 DNAJB2:97 SLC4A1AP:300
FEVR_CTNBN1_TARGETS_UP	0.06481048	600	7.198e-08	1.556e-05	ATP6V0A2:7 LPN1:50 CRY2:53 MTPP:55 LPN2:72 ITGA3:110
MARSON_BOUND_BY_FOXP3_STIMULATED	-0.05382684	855	1.212e-07	2.537e-05	BRCA1:11 TLCD2:38 MRPL42:40 SCFD2:59 FBXW1:61 DNAJC2:75
MEISSNER_NPC_HCP_WITH_H3K4ME2_AND_H3K27M	0.08738784	308	1.466e-07	2.972e-05	ST6GAL2:9 SLC18A3:84 ITGA3:110 SLC30A10:118 CACNA11:224 PTGFRN:237
REACTOME_NEURONAL_SYSTEM	0.08010365	352	2.697e-07	5.302e-05	KCNQ3:45 SLC18A3:84 CACNA18:88 GABRB3:104 KCNF1:158 KCNK18:188
KEGG_PARKINSONS_DISEASE	-0.15547938	89	4.072e-07	7.768e-05	NDUFS3:30 NDUFB8:45 NDUFB10:57 NDUFS5:72 ATP5PD:118 ATP5PB:225
RODRIGUES_THYROID_CARCINOMA_ANAPLASTIC_U	-0.06121557	577	5.945e-07	1.102e-04	VCAN:31 MRPL42:40 SSX2IP:43 KLHL5:79 PNPT1:96 SYTL2:160
REACTOME_TRANSPORT_OF_INORGANIC_CATIONS_	0.14371947	101	6.161e-07	1.110e-04	SLC5A8:145 SLC38A3:291 SLC4A1:300 SLC1A1:381 SLC26A4:401
KEGG_NEUROACTIVE_LIGAND_RECEPTOR_INTERAC	0.08965361	260	6.931e-07	1.215e-04	P2RY4:11 P2RX2:54 GABRB3:104 ADRB1:112 ADORA1:225 GPR35:272
REACTOME_RNA_POLYMERASE_I_PROMOTER_ESCAP	-0.21272659	45	7.990e-07	1.364e-04	H2BC21:15 RNR3:169 POLR1C:913 H3C13:1060 H4C2:1067 GTF2H3:1542
REACTOME_NEGATIVE_EPIGENETIC_REGULATION_	-0.18267983	60	9.998e-07	1.663e-04	H2BC21:15 BAZ2A:390 RRP8:423 DNMT3B:568 TTF1:675 DNMT1:904
REACTOME_DEPOSITION_OF_NEW_CENPA_CONTAIN	-0.22253229	40	1.124e-06	1.803e-04	H2BC21:15 CENPT:88 CENPC:128 M1S18B1:193 H4C2:1067 CENPO:1400

DisGeNET Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Mitochondrial Diseases	-0.11212190	335	2.398e-12	2.351e-08	LRPPRC:1 TMEM126A:5 NDUFS3:30 NDUFB8:45 SARS2:47 NDUFB10:57
Increased CSF lactate	-0.26044873	53	5.619e-11	2.755e-07	LRPPRC:1 NDUFS3:30 TMEM126B:65 NDUFA11:78 TRMT10C:104 NDUFAF4:132
MITOCHONDRIAL COMPLEX I DEFICIENCY	-0.30850764	26	2.532e-08	1.687e-04	NDUFS3:30 TMEM126B:65 NDUFA11:78 NDUFAF4:132 TIMMDC1:465 NDUFAF1:616
NADH:(Q)Oxidoreductase deficiency	-0.31177390	25	6.881e-08	1.687e-04	NDUFS3:30 NDUFB10:57 TMEM126B:65 NDUFA11:78 NDUFAF4:132 TIMMDC1:465
Hypertensive disease	0.04936010	1071	9.233e-08	1.811e-04	SORL1:12 NTSE:17 OFD1:32 LPN1:50 ATG7:51 CRY2:53
Lactic acidemia	-0.13260921	119	6.159e-07	1.007e-03	LRPPRC:1 NDUFS3:30 TMEM126B:65 NDUFA11:78 TRMT10C:104 SFXN4:114
Nicotinamide adenine dinucleotide coenzy	-0.22808691	39	8.389e-07	1.175e-03	NDUFS3:30 NDUFA11:78 NDUFAF4:132 RARS2:192 NDUFA12:259 HSPB3:312
Acute necrotizing encephalopathy	-0.32247993	19	1.143e-06	1.401e-03	NDUFS3:30 TMEM126B:65 NDUFA11:78 NDUFAF4:132 TIMMDC1:465 NDUFAF1:616
Progressive macrocephaly	-0.29542621	22	1.622e-06	1.768e-03	NDUFS3:30 TMEM126B:65 NDUFA11:78 NDUFAF4:132 TIMMDC1:465 NDUFAF1:616
Acidosis, Lactic	-0.10985993	146	4.871e-06	4.342e-03	LRPPRC:1 NDUFS3:30 TMEM126B:65 NDUFA11:78 TRMT10C:104 SFXN4:114
Increased serum lactate	-0.13634263	95	4.538e-06	4.342e-03	LRPPRC:1 NDUFS3:30 TRMT10C:104 SFXN4:114 TMEM70:191 RARS2:192
Diabetes Mellitus, Experimental	0.13085320	100	6.366e-06	4.458e-03	PKD4:223 GCK:286 SOD1:353 PC:373 S100A6:444 NOS3:711
Alloxan Diabetes	0.13085320	100	6.366e-06	4.458e-03	PKD4:223 GCK:286 SOD1:353 PC:373 S100A6:444 NOS3:711
Streptozotocin Diabetes	0.13085320	100	6.366e-06	4.458e-03	PKD4:223 GCK:286 SOD1:353 PC:373 S100A6:444 NOS3:711
Abnormal mitochondria in muscle tissue	-0.25633583	25	9.204e-06	6.016e-03	NDUFS3:30 NDUFB10:57 TMEM126B:65 NDUFA11:78 NDUFAF4:132 TIMMDC1:465
Increased hepatocellular lipid droplets	-0.40980081	9	2.142e-05	1.313e-02	LRPPRC:1 COA7:252 COA5:508 FASTKD2:739 COX20:1334 SCOL1:1856
Schizophrenia	0.03354316	1513	2.374e-05	1.369e-02	RGS9:3 ASAP1:5 NTRAK1:10 SORL1:12 MROH2A:27 KCNQ3:45
Essential Hypertension	0.08053421	228	3.016e-05	1.603e-02	TRPV6:25 ADRB1:112 ADD1:157 ADORA1:225 GCK:286 HSPA1L:323
Impaired exercise tolerance	-0.14732145	67	3.107e-05	1.603e-02	NDUFS3:30 DNA2:63 TMEM126B:65 NDUFA11:78 NDUFAF4:132 COA7:252
Hypertrichosis	-0.17559475	46	3.830e-05	1.878e-02	NDUFS3:30 BMS1:123 SETBP1:141 NDUFA12:259 NDUFA9:279 GATA1:534
Coronary Atherosclerosis	0.04868655	616	4.265e-05	1.991e-02	MTPP:55 PER2:56 ENTND1:60 UGT1A6:67 PRDM10:85 ADR